

Grey Matter: Where What's Real Gets Shaped Into What's Lived - A Perturbation-Response Architecture for Conscious Experience

Abstract

Conscious experience remains organized despite continuous changes in sensory, bodily, mnemonic, and internally generated information. This article develops Grey Matter, a theory-neutral framework that treats containment as the distributed, mechanistically heterogeneous processes by which an ongoing lived trajectory remains sufficiently bounded, stable, and coherent under perturbation. The framework does not propose a new gate or neural mechanism. Instead, it specifies a measurement architecture: researchers must declare the perturbation, the property of the lived trajectory, and the relation actually measured rather than treating behavior, report, physiology, memory, affect, control, or neural dynamics as interchangeable with phenomenality. Boundedness, stability, and coherence remain open candidate properties with explicit collapse rules. Once phenomenal presence is established, maintenance or termination, selective calibration, retention, and later reflection are treated as optional and separable downstream relations. Mature theories of attention, control, learning, memory, predictive processing, active inference, and neural state-space dynamics retain explanatory priority. Grey Matter survives only when relation typing and trajectory-level analysis add inferential or predictive value beyond those accounts.

Keywords: consciousness; phenomenology; cognitive dynamics; perturbation; measurement; learning; memory

1. The problem: identifying stability in lived experience

Psychological and neuroscientific explanation moves among several senses in which information can be said to be present for a person. An event can be causally available to sensory systems, encoded by the organism, accessible for report or discrimination, phenomenally experienced, affectively significant, motivationally recruiting, remembered, belief-changing, control-relevant, physiologically effective, or behaviorally expressed. These relations interact, but they are not interchangeable. Consciousness research distinguishes phenomenal consciousness from forms of access and report, while continuing to debate how those notions map onto measurement (Block, 1995; Lamme, 2003; Mashour et al., 2020; Tsuchiya et al., 2015). Masked stimuli can alter later processing and behavior without satisfying the same criteria for conscious access (Dehaene et al., 1998; Naccache & Dehaene, 2001). Reward research separates hedonic impact from incentive motivation (Berridge & Robinson, 2016). Cognitive-control and action-selection theories distinguish valuation, allocation, selection, and execution (Cisek, 2007; Shenhav et al., 2013). Automaticity research likewise rejects the inference that organized behavior requires continuous conscious supervision (Moors & De Houwer, 2006).

Grey Matter begins from a related systems problem. The organism is continuously changing while conscious experience usually retains enough continuity to form a temporally extended lived trajectory. Sensory conditions change, bodily states vary, memories and associations become active, goals and predictions shift, and action demands update. Some of those changes become prominent in experience, some are briefly incorporated, some reorganize the ongoing trajectory, and some affect the organism without becoming clearly lived. The scientific problem is not to postulate a hidden gate deciding what matters. It is to identify how an ongoing phenomenal trajectory responds to perturbation and which measured relation actually licenses a claim about that trajectory.

The central question is therefore: How does conscious experience remain sufficiently bounded, stable, and coherent while conditions capable of perturbing it continue to change?

The word containment names this stabilization problem and, only where justified, a cross-mechanism family of processes that contribute to solving it. It does not name a single cause. The framework separates novelty ownership from evidentiary use: many facts needed to motivate and constrain the proposal are mature science, and their being precedented is a reason to credit and use them rather than to erase them. Grey Matter claims novelty, if any, only for the integrated identification architecture that links a declared perturbation to a declared lived-trajectory property using relation-typed evidence and mandatory comparison against mature alternatives. A further distinction is essential: defining a state as lived does not require that it cause anything afterward, but this theory-neutral definition

does not imply that conscious phenomenality is an inert display. Once phenomenal presence has been independently established, whether and how that lived state contributes to what happens next becomes an empirical question.

Grey Matter is intentionally broad at the level of architecture and narrow at the level of inference. It is offered as a minimally precommitted reference architecture - a blueprint for organizing later, more constrained studies - rather than as a completed lower-level model. The architecture leaves open which biological, developmental, metabolic, environmental, learned, pharmacological, social, or task-specific constraints will prove relevant in a given domain. That openness is not permission to absorb any result. A concrete application must still declare the perturbation, the target trajectory property, the relation actually measured, the bridge to phenomenality when such a claim is made, the mature competing explanations, and the conditions under which the higher-level containment interpretation loses. Later work can therefore progressively constrain the reference case without treating each newly specified restriction as something the original paper had already established.

2. Relational measurement architecture

To keep the framework aligned with established literatures, Grey Matter does not posit a generic weighting variable, a generic control-leverage variable, a novel consequence-profile vector, or a novel body/world/model taxonomy. Instead, the surviving relational labels are bookkeeping devices that keep the measurement target visible. Their purpose is to prevent one demonstrated relation from silently standing in for another.

2.1 Causal exposure (E_x)

For a specified referent x , E_x denotes that x is instantiated in a way capable of causally impinging on the organism under the declared boundary. External existence alone is insufficient: a signal never encountered, outside a receptor's useful range, or otherwise isolated from the relevant system should not be treated as equivalent to an input capable of altering organismic dynamics. E_x is a bookkeeping boundary, not a theory of perception, and it does not imply encoding, conscious access, phenomenality, or downstream effect.

2.2 Organismic encoding or sensitivity (R_x)

R_x denotes evidence that some organismic state covaries with, carries information about, or functionally depends on x under an explicit measurement model. The wording deliberately avoids a strong generic commitment to internal representation, whose empirical and philosophical uses are heterogeneous (Poldrack, 2021). R_x can be demonstrated without establishing A_x or lived phenomenality.

2.3 Operational evidence of conscious access (A_x)

A_x denotes evidence that x is consciously accessible under a declared criterion such as immediate report, discrimination, confidence, or another validated measure. It is not synonymous with phenomenal consciousness. Report requirements introduce decision, memory, and response processes, motivating no-report approaches and continuing debate about what an operational measure licenses researchers to infer about experience (Tsuchiya et al., 2015). The distinction between phenomenal and access consciousness likewise remains theoretically influential and empirically contested (Block, 1995; Mashour et al., 2020).

2.4 Lived: phenomenal presence

Lived is retained as a plain-language label for phenomenal presence: there is something it is like for the organism to undergo the state. It is not defined by memory, control, report, action, or any other downstream consequence. Causal participation is deliberately excluded from the definition because building efficacy into phenomenality would make later tests of phenomenality's consequences circular.

2.4.1 Possible causal roles after phenomenality is established

The definitional restriction is not a claim that phenomenality is causally inert. It means only that causal participation cannot be used as the criterion by which a state qualifies as lived. After phenomenal presence is identified under an explicit measurement bridge, the lived state may itself become one input among others to subsequent selection,

maintenance, learning, memory, or interpretation. Whether it does so, and by which mechanism, is a separate empirical question. A null downstream effect therefore does not revoke phenomenality, and phenomenality does not guarantee any particular downstream effect.

Four possible downstream roles are distinguished. First, a lived state may contribute to maintenance or termination: preserving the current state, leaving it, interrupting it, or allowing it to continue. Second, continued experience may support learning and selective calibration about which cues should acquire more or less future influence. Third, an episode may affect encoding, consolidation, association, or later retrieval accessibility. Fourth, a later reflective episode may reinterpret or retrieve aspects of the earlier event and thereby produce additional learning or reorganization. These roles are analytically separable. None is included in the definition of lived, and none is assumed to occur simply because a state was phenomenally present.

2.5 Consequence-specific outcomes (C_x)

C_x remains only an indexed outcome placeholder. The consequence class must be named: autonomic response, reported affect, hedonic evaluation, incentive motivation, attentional priority, memory encoding, belief revision, control allocation, action selection, overt behavior, state maintenance or termination, later retrieval, or another explicitly operationalized outcome. Generic variables W_x and L_x are not used: W_x would over-compress salience, value, incentive salience, attentional priority, precision, and felt significance into one quantity, while L_x would overlap mature control-allocation and action-selection constructs (Anderson, 2013; Berridge & Robinson, 2016; Feldman & Friston, 2010; Rust & Cohen, 2022; Cisek, 2007; Shenhav et al., 2013). Those functions remain scientifically important, but they should be named and measured with the established constructs appropriate to the domain. Maintenance, retention, and later reflection are treated the same way: as named relations or later episodes, not as hidden generic weights or proof that the original state was lived.

No generic consequence-profile vector is required. Measuring several consequence classes for the same referent is useful ordinary multivariate practice rather than a new latent object. The substantive point survives without proprietary notation: when a scientific inference depends on whether x was accessible, affective, motivational, memorable, control-relevant, physiologically effective, or behaviorally expressed, those relations should be measured separately rather than collapsed into one global state of "having" x.

The same discipline applies to causal claims about lived experience. The relevant question is not whether phenomenality is generally causal or epiphenomenal in the abstract. It is whether, in a declared episode, evidence supports a bridge from phenomenal presence to a specified subsequent relation. A lived state can be followed by maintenance without later memory, by later memory without overt action, by reflection without immediate behavioral change, or by none of these. The framework treats those possibilities as measurements to distinguish rather than consequences built into consciousness by definition.

2.6 Containment-identification proposition

A containment claim is identified only when a declared perturbation is linked to a declared property of the lived trajectory using evidence typed to the relation that bears on that property. Stability in behavior, access/report, physiology, memory, affect, motivation, or control cannot substitute for stability of the lived trajectory unless an independently justified bridge is supplied.

This proposition turns relation-specific non-entailment from a generic methodological caution into the measurement-identification layer for containment. A study does not need to measure every possible relation. It must measure the relations necessary for the inference it actually makes and state which unmeasured relations are not being inferred.

2.7 Worked inference: why stable behavior does not identify stable experience

Consider an abrupt context shift in which a participant continues to perform a task successfully. A naive inference would be that the participant's conscious trajectory remained stable and containment therefore succeeded. That conclusion is not identified by the behavioral endpoint. The same performance could coexist with substantial experiential disruption that is compensated by automatized task structure, effortful control, or delayed reorganization. Conversely, a behavioral error after the same perturbation would not by itself establish experiential

fragmentation; the conscious trajectory could remain continuous while motor execution, control allocation, or another downstream process changed.

The experiment must therefore declare the target property and the evidential bridge. If the target is stability of lived experience, a phenomenology-facing measure or converging proxy must bear on continuity, displacement, or recovery of the trajectory. Behavior and physiology can be measured simultaneously, but they retain their own relation types. The containment inference comes from the declared mapping between perturbation and trajectory evidence, not from the extremity or apparent success of the outward response.

3. Established dissociations that make relation typing necessary

The relation audit is not supported by one novel experiment. It is supported by convergence across mature literatures showing that different relations to the same or related information can dissociate. Using these examples does not claim ownership of the component findings; their role is evidentiary. They explain why a containment study cannot use one convenient readout as a universal proxy for the lived trajectory.

3.1 Influence without demonstrated conscious access

Masked-priming studies provide a direct reason not to equate organismic influence with conscious access. Dehaene et al. (1998) reported neural and behavioral effects of masked semantic primes, and Naccache and Dehaene (2001) used novel unseen primes in a number-comparison paradigm to argue against a simple learned stimulus-response mapping account. These findings do not establish unlimited unconscious cognition, and their interpretation depends on the adequacy of awareness measures. They do establish the narrower measurement point Grey Matter needs: evidence that x changes processing or behavior is not, by itself, evidence that x was consciously accessible or phenomenally present.

Automaticity provides a related family of cases. Moors and De Houwer (2006) emphasize that automaticity is multidimensional: a process can be efficient, unintentional, difficult to control, goal-independent, or outside awareness in different combinations. A familiar procedure may therefore be organized successfully without every information-bearing component becoming a distinct object of conscious supervision. Grey Matter does not call such processing "contained" when an established automaticity construct explains it; it uses the precedent to block an invalid reverse inference from successful behavior to continuous conscious representation.

3.2 Cognitive identification without expected affective consequence

The relation between explicit recognition and affective or autonomic consequence is also nonunitary. Ellis et al. (1997) reported reduced autonomic responses to familiar faces in Capgras delusion despite preserved overt identification, supporting models in which recognition and autonomic familiarity responses depend on partially separable processes. Shamay-Tsoory et al. (2009) reported a lesion-based double dissociation between cognitive and emotional components of empathy. Neither finding implies that cognition and affect are generally independent. Together they show that identifying or understanding a person, event, or proposition does not logically guarantee a particular affective consequence.

This matters for the paper's ordinary-language examples. "If you knew, you would care" may be a reasonable prediction in a particular context, but it is an empirical prediction, not a definition of knowledge. A containment study concerned with whether an event became lived cannot infer the strength of its affective consequence merely from semantic report, nor can weak affect be used to infer absence of semantic or conscious access.

3.3 Felt or hedonic state, motivation, control, and action

Reward and control literatures similarly resist a single generic "importance" variable. Berridge and Robinson (2016) distinguish hedonic "liking" from incentive-salience "wanting." Shenhav et al. (2013) model control allocation as depending on expected benefits, costs, and efficacy, while Cisek (2007) treats action specification and selection as interacting competitive processes. These established constructs occupy much of the space once compressed into W_x and L_x . Their precedent is precisely why those generic variables are not retained.

The explanatory consequence, however, remains important. A state can be consciously felt without recruiting the expected action; motivational value can be present while another action wins selection; a selected policy can fail at execution; and an overt behavior can occur under constraints that do not reveal how strongly the triggering content was experienced. A vivid hunger state without eating, or consciously reported fear without avoidance, are therefore useful thought experiments because they force the experimenter to ask which of these relations actually changed. They are not offered as evidence for a special containment mechanism.

A further possibility is that no overt change is itself part of the selected outcome. Response-inhibition research shows that prepotent actions can be actively stopped or paused, including under internally generated goals (Aron et al., 2014). That literature does not establish a general phenomenological theory of state maintenance, but it blocks the inference that inaction is necessarily absence of control. A person who remains still, stays in an interaction, or deliberately does not terminate a state may be expressing control through maintenance rather than through an observable transition. Whether that is what occurred must be measured; stillness alone cannot establish agency.

3.4 Behavior without continuous conscious supervision

A familiar route, skilled motor routine, or well-practiced task can remain organized while conscious attention is partly occupied elsewhere. The correct conclusion is not that skilled behavior is unconscious. Skilled performance can recruit awareness when errors occur, circumstances change, or deliberate adjustment becomes necessary. The narrower point is that behavioral success does not demonstrate continuous conscious supervision of every component. This is one reason the containment-identification proposition cannot use performance stability as a transparent measure of experiential stability.

3.5 Conscious experience and later recollection

Later memory is another downstream relation, not a retrospective meter of phenomenality. Jacoby's (1991) process-dissociation framework demonstrates the broader methodological need to separate contributions that can jointly determine memory performance, while consciousness research emphasizes that report itself adds cognitive operations (Tsuchiya et al., 2015). The literature does not supply a perfect memory-independent measure of phenomenality. It does support a constraint: weak later recollection should not be treated as proof that an earlier interval was not consciously experienced.

A familiar-travel example makes the inferential problem concrete. A person can complete a highly practiced journey and later recall little of it. That endpoint does not establish whether the route was wholly absent from experience, intermittently available, or simply poorly encoded for later episodic retrieval. Behavior, online experience, and later memory require separate evidence. The example is explanatory rather than evidentiary; the evidentiary support comes from the established separation of memory/report processes from other conscious-access measures.

3.6 Maintenance, retention, and reflection are distinct consequences

Maintenance and termination should be measured as outcomes in their own right. Phenomenal discomfort does not logically entail escape, and positive phenomenal value does not logically entail approach or overt action. One person may experience fear while remaining in place; another may experience a highly valued quiet interaction and intentionally preserve it by doing very little. These are explanatory examples, not evidence that phenomenality alone caused the maintained state. The empirical question is whether the lived state contributed to a control process that favored continuation over termination, after competing explanations such as automaticity, task demand, inhibition, reward, avoidance, or social constraint are measured.

Retention is another separable consequence. Emotional arousal can modulate consolidation of lasting memories (McGaugh, 2015), and reward motivation can prioritize information for encoding and consolidation (Miendlarzewska et al., 2016). These literatures support the general proposition that later accessibility depends on established memory and motivational mechanisms, not on a generic memory-value variable. A lived episode can therefore alter what is easier to retrieve later, but later retrieval is neither guaranteed by phenomenality nor required for the earlier episode to have been lived.

Reflection is later still. Reappraisal research shows that changing the interpretation of an event can alter subsequent emotional response (Ochsner & Gross, 2005), and retrieval research shows that retrieving earlier material can

influence later learning and future retrieval (Chan et al., 2018; Kensinger & Ford, 2020). Grey Matter therefore treats reflection as a new causal episode: original lived episode -> possible retention -> later retrieval or reflection -> possible reorganization. The later episode can change associations, contextual meaning, expectations, or future cueing without being a transparent reconstruction of the original phenomenology.

3.7 Branching rather than mandatory serial architecture

Taken together, the dissociations defeat a mandatory universal story in which information must first become consciously available, then acquire one generic weight, then gain one generic form of control, and finally produce behavior. Influence can occur without demonstrated access; conscious or semantic presence can coexist with weak affective or motivational consequence; control and action selection have their own structure; organized behavior can rely on automatized processing; and later memory can diverge from earlier access or experience. The architecture therefore has to branch at the level of inference even if a particular domain later supports a more specific serial model.

Grey Matter does not claim that every branch is available in every task or that conscious access never matters. A rigorous study may find that a particular consequence requires access under its boundary. Another may find that access is sufficient for several downstream outcomes. Those results should reduce the architecture for that domain. The only general rule is that the bridge must be earned rather than assumed.

Several thought experiments help expose this logic without pretending to supply data. A person can encounter counterevidence, accurately summarize it, and yet show little revision of a broader model; semantic uptake and belief change are then distinct measurement questions. A person can experience a valued outcome and a brief positive state that rapidly gives way to a different evaluative process; reward occurrence and reward persistence are then separate questions. These examples are useful only if they lead to explicit measures rather than retrospective labels such as denial, suppression, or low containment.

4. Defining containment

Containment is the distributed and potentially mechanistically heterogeneous set of processes by which conscious experience remains sufficiently bounded, stable, and coherent despite ongoing changes in internal and external information.

Each term constrains the proposal. Containment is not a physical container, a single anatomical gate, deliberate suppression, or a hidden causal force. It is not equivalent to attention, gating, precision, salience, inhibition, control, allostasis, or neural state-space geometry. Those are established mechanism-specific or structural constructs. They may contribute to a particular stabilization problem, and when they fully explain the phenomenon their own names should be used without adding containment as an extra cause.

"Sufficiently" is essential. The framework does not define one globally optimal amount of constraint. A trajectory can be too susceptible to irrelevant perturbation, but it can also be too rigid to incorporate information that should update it. The target is a viable dynamical range under a declared organism, task, state, and timescale, not a monotonic rule that more containment is healthier.

That viable range can also be history-sensitive without behaving like one adjustable threshold. Calibration means that prior experience may change the future influence of particular cues, contexts, or transition classes. Some perturbations may acquire less power to redirect or terminate the trajectory, while others acquire more because they prove predictive, diagnostic, or consequential. A signal can remain fully phenomenally present while its authority over the next transition changes. The breakwater metaphor should therefore not be translated into a single wall height; any learning claim must be expressed in the established mechanism-specific terms supported by the data.

5. Candidate trajectory properties: boundedness, stability, and coherence

Boundedness, stability, and coherence are treated as candidate properties of the lived trajectory rather than as synonyms. Their conceptual distinction is useful, but their empirical independence remains OPEN. They are not declared orthogonal axes, and they must be merged or reduced if sensitive measurements fail to dissociate them.

5.1 Boundedness

Boundedness concerns selective incorporation: not every causally available or organismically encoded change acquires equal power to enter, persist within, or reorganize the lived trajectory. Boundedness is not impermeability. A highly responsive conscious system can remain bounded if perturbations influence it selectively rather than indiscriminately. History may change that selectivity in cue- and context-specific ways: one class of input may become less able to redirect the trajectory while another becomes more influential. Candidate indicators include susceptibility to task-irrelevant interruption, the probability that a distractor becomes reportably incorporated, and the degree to which a declared perturbation redirects or terminates subsequent experiential content. These are proxies whose evidential role must be declared rather than a universal boundedness meter.

5.2 Stability

Stability concerns persistence, resistance to displacement, and recovery across time. Stability is not stasis. A trajectory can change rapidly while remaining stable in the relevant sense if transitions preserve enough continuity that perturbations do not repeatedly reset or fragment its organization. Candidate indicators include dwell time in a declared task or thought state, recovery time after interruption, transition rate under matched perturbations, and structured reports of subjective discontinuity. Neural state-space stability can provide a mechanistic or structural comparator but is not silently equated with phenomenal stability.

5.3 Coherence

Coherence concerns structured mutual constraint among experiential contents and transitions sufficient to constitute an ongoing trajectory under the boundary being studied. It is deliberately not defined by logical truth, emotional calm, clinical normality, social desirability, or task success. A vivid dream, intense grief, or unusual altered state can be coherent; calm behavior can coexist with a poorly integrated subjective sequence. Candidate indicators include consistency of immediate source/context judgments, preservation of task-model relations across transitions, and structured continuity reports. Each requires a theory and measurement bridge to phenomenality.

5.4 Pairwise dissociation tests

Boundedness and stability can dissociate in principle. High boundedness with low stability could occur when irrelevant external input is effectively excluded but internally generated contents transition rapidly, leaving the trajectory unstable despite low external capture. Low boundedness with high stability could occur when many perturbations are incorporated into experience but are integrated into a persistent organizing theme or task model rather than displacing it.

Stability and coherence can also dissociate. High stability with low coherence could appear as a persistent stream whose contents remain poorly related in source or context: the trajectory endures but its internal relations are disorganized. Low stability with high coherence could appear as rapid switching among episodes that are each internally structured and source-consistent; persistence is low even though local organization is high.

Coherence and boundedness can likewise separate. High coherence with low boundedness could describe a trajectory that readily incorporates many incoming events yet integrates them into a structured sequence. Low coherence with high boundedness could describe a trajectory resistant to external capture but internally disorganized or source-confused. These are candidate quadrants, not established syndromes. If experiments cannot produce or detect such dissociations, the dimensionality should be reduced rather than protected by definition.

5.5 Measure-to-construct map

Candidate measures should be attached to a target property rather than listed generically. Transition entropy is primarily a stability indicator when it quantifies how often or unpredictably the trajectory changes, but it may also bear on coherence if transitions are modeled relationally. Subjective discontinuity is a phenomenology-facing stability proxy when it asks about breaks or resets in the experienced sequence. Interruption susceptibility and task-model persistence primarily probe boundedness and stability: they estimate how readily irrelevant perturbation displaces an ongoing organization and how well that organization recovers. Source judgments primarily probe

coherence when the hypothesis concerns preservation of source/context relations. None of these is direct access to phenomenality, and converging measures are preferable when the inference is strong.

6. Sources of perturbation and mechanistic pluralism

Grey Matter does not propose a novel body/world/model taxonomy. The relevant conceptual space is already covered by established work on interoception, external/internal attention, perceptual decoupling, and self-generated thought (Quadt et al., 2018; Chun et al., 2011; Smallwood & Schooler, 2015; Christoff et al., 2016). Those literatures remain useful for explaining source and coupling differences without introducing proprietary notation.

6.1 Established source and coupling distinctions

Interoception studies body-to-brain signaling, neural encoding of bodily state, and the relationships between bodily information, cognition, behavior, and consciously accessible sensation (Quadt et al., 2018). Attention research distinguishes attention directed toward external sensory information from attention directed toward working memory, long-term memory, task rules, and other internally represented information (Chun et al., 2011). Mind-wandering and spontaneous-thought research examines perceptual decoupling and self-generated cognition whose continuation can depend less on immediate external input (Smallwood & Schooler, 2015; Christoff et al., 2016).

These established distinctions matter because perturbations can originate in or recruit different channels without implying a single new source architecture. A change in pain, heartbeat, hunger, or arousal can perturb experience through bodily/interoceptive pathways. A sudden environmental or social event can perturb it through exteroceptive channels. A memory, prediction, image, association, or ongoing problem representation can perturb it from internally generated activity. The categories interact continuously and should not be treated as mutually exclusive modes.

A skilled-work example illustrates the point without introducing a new source vector. An experienced programmer can sustain a reasoning sequence from an active task model while most surrounding visual information remains irrelevant to the next meaningful step. When a specific missing datum becomes necessary, the person samples the environment, extracts the needed syntax, and returns to the ongoing problem. This is not evidence for a proprietary "model-sustained" state. It is an explanatory case consistent with established internal-attention and self-generated-thought literatures: the relevance of fresh environmental information can change dynamically while the conscious trajectory remains organized.

6.2 No inner chooser is required

Terms such as filtering, gating, blocking, permission, and containment can accidentally introduce a homunculus. No inner chooser is required. Distributed competition, state-dependent gain or precision, attentional allocation, action-selection dynamics, learned routing, recurrent processing, automaticity, and source-monitoring processes already provide non-homuncular ways for some information to exert more influence than other information (Cisek, 2007; Feldman & Friston, 2010; Moors & De Houwer, 2006; Shenhav et al., 2013).

The distinction between description and mechanism is therefore strict. "The perturbation was consciously accessible but had little motivational effect" describes a measured relation pattern. "The information was inhibited by a defensive process" proposes a mechanism and requires separate evidence. "The task continued because learned automatic organization reduced conscious supervision" is a different mechanistic hypothesis. Grey Matter supplies no permission to replace those alternatives with an unseen containment process.

6.3 History, adaptation, and context

History dependence is established background rather than a Grey Matter invention. Learning history can alter later attentional priority (Anderson et al., 2011; Anderson, 2013), automaticity changes with practice (Moors & De Houwer, 2006), and emotion-regulation outcomes can depend on person-by-situation factors such as controllability (Troy et al., 2013). Repeated psychosocial stress shows response-system-specific habituation rather than a single generalized resilience effect (Barthel et al., 2025). Fear-learning and exposure literatures add a particularly useful constraint: extinction is not well described as simply erasing the original association or globally turning fear down. Inhibitory-learning approaches emphasize expectancy violation, retrieval, stimulus variability, and multiple contexts,

while conditioning research shows that context and retrieval can determine whether an extinguished response returns (Bouton & Moody, 2004; Craske et al., 2014, 2022).

The containment consequence is a selective-calibration hypothesis rather than a toughness or desensitization claim. Prior experience may alter how a particular cue or transition class is incorporated, tolerated, used, or recovered from while leaving another class unchanged or even increasing its future influence. Human discrimination training can reshape fear-generalization gradients, providing direct precedent that learning can refine responses across similar stimuli rather than merely lowering a global response level (Dunsmoor & LaBar, 2013). A pattern that is useful under one context can become costly if it persists after the relevant environment changes. Adaptiveness therefore cannot be part of the definition of containment; it is an outcome evaluated relative to task, state, timescale, and environment.

6.4 Maintained states can create additional sampling opportunities

A maintained state can sometimes extend the interval over which the organism samples what actually follows a perturbation. In an aversive case, remaining in a non-catastrophic situation long enough to observe that an expected outcome does not occur can generate prediction error or inhibitory learning; in a discrimination case, additional sampling can help separate diagnostic cues from merely similar ones (Craske et al., 2014, 2022; Dunsmoor & LaBar, 2013). This is supporting precedent for selective learning, not evidence that all maintained discomfort is useful or that exposure therapy is an instance of Grey Matter.

The same architecture is not restricted to threat. Positive experience can also alter later priorities, associations, and memory. Reward motivation can preferentially support encoding and consolidation of valuable information (Miendlarzewska et al., 2016). A quiet interaction, absorbed activity, or other valued state may therefore be maintained without much overt behavior and may later become accessible or worth returning to. Grey Matter treats those as possible consequences mediated by established reward, memory, and control mechanisms; it does not collapse positive and negative learning into one process.

7. Prior art and the boundary of the contribution

The central components of Grey Matter have substantial prior art. Christoff et al. (2016) characterize spontaneous thought in terms of constraints on mental-state contents and transitions. Zamani, Carhart-Harris, and Christoff (2022) directly discuss stability and variability of thought and conscious experience and synthesize deliberate and automatic constraints, including psychedelic alterations. More recently, Sheth et al. (2026) used experience sampling across four experiments to show that freely moving and deliberately directed thought were distinguishable reported dimensions across experimental contexts, supporting the feasibility of probing differentiated first-person thought dynamics without establishing Grey Matter's candidate dimensions. McEwen's allostatic framework precedents dynamic stability through change at a broader organismic level (McEwen, 1998). Smith (2026) synthesizes work in which neural states associated with conscious experience occupy restricted, structured, dynamically accessible state spaces with stability properties under perturbation.

Grey Matter therefore does not claim that conscious states are constrained, that thought has stability and variability, that neural state spaces can be restricted or structured, that biological systems maintain stability through change, or that attention, prediction, control, gating, and source monitoring alter influence. Those propositions are PRECEDENTED. Nor does it claim that dissociation logic or construct-validity discipline is new; process dissociation, cognitive ontology, no-report methods, and domain-specific dissociation literatures already instantiate that logic (Jacoby, 1991; Poldrack et al., 2011; Poldrack & Yarkoni, 2016).

Closer measurement precedent comes from work that treats consciousness measures themselves as theoretically committed and potentially non-equivalent. Seth et al. (2008) review behavioral and neurophysiological measures of consciousness, showing that different measures can presuppose different theories and can yield conflicting classifications; Gamez (2014) develops a theory-neutral framework that makes first-person reporting and the assumptions required to link reports, consciousness, and physical correlates explicit. These works PRECEDENT the general consciousness-measurement problem and the need to state what a measure licenses; Grey Matter does not claim to invent that problem. Its OPEN residual is narrower: the integrated perturbation-response architecture that couples a declared perturbation and lived-trajectory property to relation-typed evidence, a justified phenomenology

bridge where needed, mature competing explanations, optional downstream relations, and explicit loss or collapse conditions.

Contemporary active-inference theories provide a still broader integrative competitor. Whyte et al. (2026) derive a minimal theory of consciousness from common features of active-inference models, while Laukkonen et al. (2025) propose a computational account organized around a world model, inferential competition, epistemic depth, and precision control. These frameworks can potentially absorb many local accounts of prediction, action, learning, selection, and altered conscious dynamics. Grey Matter does not compete with them as a computational mechanism. Its residual claim is narrower: a measurement architecture for deciding what a perturbation changed in the lived trajectory and which relation-specific observations license that inference. If active inference, together with a validated bridge to phenomenality, supplies the same relation typing, downstream branch separation, and loss discipline, the residual Grey Matter contribution would be SUBSUMED.

The residual object is narrower: a phenomenology-facing perturbation-response measurement architecture in which the lived trajectory is the target, perturbations are parameterized, each observation is relation-typed, and the higher-level construct must lose whenever mature lower-dimensional or state-space models fully explain the effect. Smith (2026) is not a rival to be displaced; it is a comparator at a different analytical level. If neural state-space organization plus a validated bridge to experience supplies the same integrated identification architecture, the residual Grey Matter contribution is SUBSUMED and should be narrowed or abandoned.

8. A structured perturbation description

A perturbation is a declared change with the potential to alter the current lived trajectory. The framework does not assume one universal perturbation scalar. The organization below separates properties of the change from what the organism expects and from the state in which the change is received.

8.1 Intrinsic perturbation properties

Magnitude or endpoint displacement concerns how far the relevant information or state moves from its starting condition. Rate concerns how quickly that change occurs. Duration or temporal extent concerns whether it is brief, prolonged, repeated, or sustained. These properties can interact: duration can alter integrated exposure and create opportunities for adaptation, while rapid rate can increase abruptness even when the final endpoint is matched.

8.2 Expectation and history variables

Predictability concerns whether the transition is expected under explicit instruction or the organism's current model. History-relative familiarity concerns prior exposure to structurally similar transitions. They may correlate, but they are not identical by definition. A familiar perturbation can be explicitly unpredictable on a given trial, and a novel perturbation can be explicitly announced. The distinction remains OPEN until experiments show incremental value beyond learned expectation, surprise, and related constructs.

8.3 Source or channel descriptor

Source describes where the relevant perturbing information enters the analysis: environmental or social, bodily/interoceptive, internally generated or model-related, mnemonic/associative, or another declared class. Source is not automatically an independent causal dimension. Different sources may simply recruit different established mechanisms. Its value is descriptive unless matched experiments show source-specific effects after salience, arousal, predictability, and mechanism are controlled.

8.4 Organism-state moderators

Current arousal, fatigue, task load, goals, pharmacological state, and related organism conditions are moderators rather than intrinsic properties of the perturbation. The same nominal event can therefore have different trajectory effects in different states without implying that the external change itself became larger. Any state effect should first be tested against mature accounts of arousal, vigilance, fatigue, attention, or control.

9. Discriminating predictions

The perturbation scaffold earns scientific value only by creating comparisons that can lose against simpler alternatives. The following predictions are prospective rather than established findings.

9.1 Rate versus matched endpoint

Hold starting state, final or endpoint displacement, and explicit forewarning as constant as feasible while varying transition rate. If rate contributes independently to trajectory stability, the rapid-transition condition should show greater displacement, slower recovery, higher interruption susceptibility, greater subjective discontinuity, or another preregistered stability measure despite the matched endpoint. A simpler account wins if endpoint magnitude, surprise, arousal, attentional capture, or a mature state-space model fully explains the difference. Failure to observe a rate increment should remove or narrow the independent rate claim.

9.2 Familiarity versus matched explicit predictability

Match explicit predictive information about a perturbation while comparing a history-familiar transition with a structurally similar novel transition. If history-relative familiarity contributes beyond explicit predictability, the familiar condition should show lower perturbation capture, faster recovery, or another preregistered trajectory effect. The claim loses if explicit predictability, arousal, learning strength, or a more specific mechanism fully explains the difference. This comparison operationalizes the idea that prior experience can calibrate response without converting experience into a global resilience scalar.

9.3 Optional moderator and source tests

A third comparison can hold the perturbation fixed while varying a declared organism-state moderator, such as rested versus fatigued or low versus high task load. Grey Matter predicts moderation, not a universal direction. If the effect is completely captured by vigilance or control-demand models, containment is unnecessary for that comparison.

Source can be tested only with similar discipline. For example, an experiment might compare matched interruptions carried by external sensory events versus internally generated reminders while controlling expectedness and subjective salience as well as possible. A source effect is interesting only if it cannot be reduced to the mature mechanism that differs between those conditions. The purpose is to ask whether source helps organize the perturbation-response problem, not to resurrect a body/world/model axis.

10. Nested competing models and loss conditions

Containment is scientifically useful only if it survives comparison with simpler or more mature accounts. The competitor should be specified before the critical test because different alternatives explain different apparent "containment" effects.

Arousal or intensity can explain large transient disruptions after salient events. Attentional-capture models can explain rapid recruitment by task-irrelevant stimuli. Surprise, prediction error, precision, or volatility can explain effects of unexpected transitions. Capacity or working-memory interference can explain failure to maintain a task representation under competing load. Control-demand models can explain costly stabilization when continued performance requires additional effort. Dynamic-thought models can explain changes in content-transition constraints. Neural state-space models can explain accessibility, restricted regions, transition structure, and recovery under perturbation. Active-inference models can jointly explain prediction, precision, action, learning, and candidate mechanisms of conscious organization (Laukkonen et al., 2025; Whyte et al., 2026). Sensory-gating and source-monitoring models can explain mechanism-specific filtering or source-confusion effects. None should be renamed as containment when it already provides the needed explanation.

The universal loss rule is: if mature lower-dimensional or neural state-space variables fully explain the perturbation-to-trajectory effect and relation typing adds no further inferential constraint, containment is scientifically unnecessary for that effect and should be dropped. The framework is not allowed to survive by redescribing whatever variance remains after each successful reduction. A legitimate higher-level role requires reproducible cross-mechanism

regularity, a measurement correction that prevents a wrong trajectory inference, or incremental prediction not supplied by the competitor.

Measurement artifact is itself a competitor. Failure to report does not prove absence of experience; failure to act can reflect environmental or motor constraints rather than low motivation; self-report can be noisy; and memory tests can miss information that was available under another retrieval condition. Null relations therefore require adequate sensitivity, converging measures, equivalence tests, Bayesian evidence, or another design appropriate to the claim. A framework built from unmeasured zeros would be unfalsifiable.

11. Boundary cases graded by evidential role

Boundary cases are useful because they can expose different parts of the proposed regime. They are not evidence for one common containment mechanism, and their evidential roles differ.

11.1 Psychedelics - phenomenology-informative and mechanism-informative

Psychedelic models provide direct precedent for altered constraint while consciousness persists. REBUS proposes reduced precision of high-level priors under serotonergic psychedelics, increasing the influence of bottom-up information (Carhart-Harris & Friston, 2019), while Zamani et al. (2022) discuss altered stability and variability of conscious experience. These findings are phenomenology-informative because subjective experience is a central outcome and mechanism-informative because predictive/precision accounts make specific claims. They do not show that all information reaches consciousness or that one containment system has failed.

11.2 Mania and psychosis - mixed phenomenology- and mechanism-informative probes

Mania is a candidate transition-dynamics edge case, not a definition of low containment. Process-oriented verbal-fluency work in bipolar disorder reports measurable changes in switching and clustering that can serve as external markers of altered cognitive dynamics (Weiner et al., 2019). Psychosis provides distinct candidate mechanisms: sensory-gating differences are reported in schizophrenia and bipolar disorder (Atagun et al., 2020), while source-monitoring meta-analyses identify systematic impairments in distinguishing internally and externally generated material (Brookwell et al., 2013; Damiani et al., 2024). These literatures are informative about candidate mechanisms and some subjective phenomena, but they do not justify collapsing mania, psychosis, and psychedelia into one failure mode.

11.3 Repeated stress - mechanism-informative and architecture-illustrating

Repeated-stress work illustrates history dependence without licensing a generic toughness construct. Barthel et al. (2025) report relatively robust HPA-axis habituation across repeated psychosocial stress while autonomic and immune responses are less uniform. That asymmetry is mechanism-informative and supports the need to specify which response relation adapts. It is architecture-illustrating for Grey Matter because it motivates history-relative familiarity and organism-state moderation, but it does not directly establish phenomenal containment.

11.4 Total solar eclipse - architecture-illustrating only

Zoo-animal responses to total solar eclipses show that abrupt environmental discontinuities can produce heterogeneous organismic and behavioral responses across species and contexts (Hartstone-Rose et al., 2020; Lazure et al., 2026). These studies are architecture-illustrating only. They support the general premise that an unusual environmental transition can reorganize behavior, but they cannot establish the structure of nonhuman phenomenal experience and therefore do not provide evidence for conscious containment as defined here.

11.5 Abrupt positive and negative events - architecture-illustrating hypotheses

The framework does not build valence into perturbation, learning, or retention. Sudden loss, threat, unexpected success, an outcome believed impossible, or another high-impact update can differ enormously in mechanism and meaning. Their role here is to motivate the hypothesis that endpoint displacement, rate, predictability, and history can matter independently of positive versus negative valence. Positive experiences can also support learning about

states, contexts, people, or activities that are worth preserving or returning to; reward-memory research provides precedent that motivational relevance can prioritize encoding and consolidation (Miendlarzewska et al., 2016). Conversely, an aversive state can remain phenomenally vivid while no escape occurs. These examples do not imply common mechanisms. They show why valence, phenomenality, persistence, action, and later accessibility must be measured separately.

12. Empirical program: from architecture to discriminating studies

A Grey Matter study begins by defining the perturbation, the target trajectory property, the measurement relations, and the competitor model. This order is intended to prevent a broad umbrella label from absorbing whatever result happens to occur. The methods used to implement the program are established dissociation, factorial, within-person, multivariate, and model-comparison methods rather than a new experimental technology (Jacoby, 1991; Shamay-Tsoory et al., 2009).

One use of the reference architecture is progressive constraint. A study may begin from the broad reference case and impose or select a declared restriction - for example, a developmental or learned history, a physiological or interoceptive state, an environmental or social structure, a pharmacological condition, memory accessibility, control capacity, source availability, or a perturbation regime defined by type, magnitude, duration, rate, predictability, or familiarity - and then ask how the perturbation-to-lived-trajectory relations change. These are examples of possible constraint classes, not established Grey Matter dimensions or predicted effects. The relevant local mechanism should be modeled first. The higher-level question is whether any reproducible trajectory-level regularity remains after that explanation is fitted. Across studies, evidence may justify subdividing, merging, restricting, or eliminating parts of the reference architecture.

12.1 Multi-relation measurement of one referent

A basic design selects one referent or perturbation and measures more than one relation in the same participant and state. Depending on the question, the design could include immediate access/report, affective response, incentive motivation, memory encoding or later recall, source judgment, control allocation, state maintenance or termination, physiology, and overt behavior. The goal is not a single containment score. It is to determine whether the relation needed for the trajectory inference moves with or apart from the convenient downstream measures. When the hypothesis concerns a causal role of phenomenality, the phenomenology-facing measure and the downstream outcome must be collected as separate variables rather than defined into one another.

If the relevant relations always move together under sensitive manipulations and one mature latent construct explains them, the higher-dimensional description should collapse for that domain. If they dissociate reliably, the measurements justify keeping those relations separate even though the dissociation method itself is established.

12.2 Hold conscious access approximately constant

One informative design holds operational access approximately constant while another relation varies. Participants might identify x with similar accuracy across conditions while affective response, motivational recruitment, memory, source confidence, or action selection differs. Such a design does not prove a unique containment process. It demonstrates why access is insufficient to identify the downstream relation and prevents a containment claim from silently substituting reportability for the trajectory property of interest.

12.3 Hold behavior approximately constant

The reverse design holds overt behavior approximately constant while testing whether internal relations differ. Two participants or two states may both complete a skilled task, yet one may require active conscious monitoring while the other relies more heavily on learned automatic organization. Two may both fail to act, yet one may report strong desire and conflict while another reports little motivational recruitment. Behavioral equivalence can therefore conceal different organizations, which is precisely the inferential problem the relation audit is intended to expose.

12.4 Within-person transitions and history

Within-person designs are especially useful because the framework predicts state-dependent response to matched perturbations. The same person can be studied across fatigue, learning, attentional demand, or other ethically and scientifically justified conditions. State comparisons must not become post hoc explanations: the moderator should be declared in advance, and the analysis should first test mature state-specific competitors.

Repeated-exposure designs can test whether adaptation is specific to the trained perturbation class. The history hypothesis predicts selective transfer rather than generic hardening. A failure to observe class-specific effects, or successful reduction to ordinary learning and expectation variables, should narrow the history component.

12.5 Source and targeted-sampling experiments

Established source distinctions can also generate experiments without introducing a new source vector. A task can vary how much continuation depends on fresh external information, interoceptive feedback, or internally maintained task content while measuring immediate experience, interruption susceptibility, and performance separately. A reasoning task that proceeds from an active internal representation but occasionally requires a specific external datum should be disproportionately disrupted by removing the required datum compared with removing irrelevant surrounding information. That result would be interpretable through established attention and task-model constructs first; its Grey Matter relevance would be to the way a declared source perturbation changes the lived trajectory, not to a new source taxonomy.

12.6 Temporal indexing as a measurement-boundary example

Temporal experience offers a narrow illustration of why process history and explicit temporal representation should not be conflated. Prospective and retrospective duration judgments rely partly on different processes (Block & Zakay, 1997). An ongoing thought can contain many internal transitions without requiring explicit clock information at each step; a later duration estimate may therefore depend on reconstruction. Grey Matter does not use this example to claim a general theory of temporal self-location. It uses it to show that later report, online representation, and the externally measured duration of a process are different relations and should be typed accordingly.

12.7 Intervention specificity

Mechanism-specific interventions provide another way to prevent umbrella explanation. A manipulation that changes attentional allocation need not change incentive motivation; an intervention affecting motivational recruitment need not change episodic memory; a source-monitoring manipulation need not change behavioral vigor. When a mature mechanism predicts a selective effect, that mechanism should receive explanatory priority. Cross-domain containment becomes useful only if the selective effects reveal a reproducible trajectory-level regularity that is not captured by simply listing the local mechanisms.

12.8 Phenomenality versus maintenance or termination

A direct test can separate lived discomfort from escape or termination. A controlled perturbation can be calibrated to produce a consciously reportable aversive state while participants retain a safe option to stop or continue. The primary variables would include a declared phenomenology-facing measure, termination latency or continuation duration, control-demand or inhibition measures, and relevant physiological indices. The prediction is not that discomfort causes staying. It is that phenomenality and termination can vary independently enough to require separate measurement. If established avoidance, response-inhibition, reward, or control-allocation models fully predict continuation, those models win and containment adds no causal explanation (Aron et al., 2014; Shenhav et al., 2013).

The positive complement can hold overt action near zero while varying experienced value. For example, two quiet maintenance intervals could require the same physical inactivity while differing in reported desire to continue, later return choice, or memory accessibility. Such a design can test whether positive phenomenal value, state maintenance, overt action, and later retention dissociate without assuming that any one is a proxy for another. Reward and memory models should be fitted before invoking a higher-level containment interpretation.

12.9 Selective calibration versus global habituation

To test selective calibration, repeated experience should be followed by probes that distinguish cue-specific or context-specific change from a global fall in response. A design can compare a trained cue, perceptually or conceptually similar cues, and diagnostic cues that genuinely predict the relevant outcome. If continued sampling produces a narrower or reorganized generalization gradient while response to diagnostic information is preserved or enhanced, the result is compatible with selective refinement. If every cue simply evokes less response, a global habituation account may be sufficient. Discrimination and inhibitory-learning literatures provide mature models for these effects and retain explanatory priority (Bouton & Moody, 2004; Craske et al., 2014, 2022; Dunsmoor & LaBar, 2013).

Context must also be tested because extinction and inhibitory retrieval can be context-sensitive. A calibration claim that appears only in the training context and disappears with renewal or reinstatement should not be described as a general restructuring of containment. Conversely, context-specificity may itself be the scientifically correct result. Grey Matter contributes only the requirement to state which future perturbation-to-trajectory relation changed; it does not replace the learning account.

12.10 Retention and reflection as later causal episodes

Retention can be tested without making memory a criterion for phenomenality by measuring the original episode online and then assessing encoding, consolidation-sensitive retention, or later retrieval separately. Emotional arousal and reward motivation can influence memory through established mechanisms (McGaugh, 2015; Miendlarzewska et al., 2016). A finding that later accessibility differs while original phenomenology is matched would support a downstream retention relation; it would not redefine what was lived.

Reflection can be tested as a later intervention. After an original episode has been measured, a subsequent session can manipulate retrieval, reinterpretation, or structured reappraisal and then test future cue response, retrieval organization, expectation, or trajectory stability. Reappraisal and retrieval literatures show that later processing can alter emotional response and future learning (Ochsner & Gross, 2005; Chan et al., 2018; Kensinger & Ford, 2020). The critical design requirement is temporal separation: evidence collected during reflection cannot be used as if it directly measured the original phenomenology.

13. Implications for psychological explanation

Several broader implications follow from the current architecture. They are not claimed as new discoveries where mature literatures already support their components. Their purpose is to clarify what follows if relation typing and perturbation-to-trajectory analysis are taken seriously.

13.1 Information possession is relational rather than unitary

Statements such as "the person knows," "the person feels," "the person wants," or "the person remembers" name different relations between content and organism. They may predict one another under familiar conditions, but they are not logical entailments. The scientific question is therefore not whether information is globally "inside the person" but which relations to that information have actually been demonstrated. Cognitive-ontology work already emphasizes that psychological constructs and tasks do not map one-to-one (Poldrack et al., 2011; Poldrack & Yarkoni, 2016); Grey Matter applies the same discipline to containment claims.

13.2 Similar behavior can mask different internal organization

Two states can arrive at the same behavioral endpoint through different internal organizations. Conversely, similar self-report can accompany different physiological or behavioral consequences. This is not merely philosophical underdetermination; it is a design problem. Whenever behavior is used as a proxy for lived stability, the study must show why that bridge is valid under the boundary being tested. Otherwise outward composure, successful performance, or behavioral failure can be mistaken for a property of experience that was never measured.

13.3 Context-dependent persistence and viable range

A stabilizing organization can be useful in one environment and costly in another without becoming a different mechanism merely because its value changed. Context-sensitive emotion-regulation findings and state-dependent cognitive effects make this general point unsurprising (Troy et al., 2013). Grey Matter uses it to reject a normative definition of containment. Persistence, flexibility, and responsiveness must be evaluated relative to the perturbation and task; "adaptive" is an outcome judgment, not a hidden property of the process.

13.4 Calibration and measurement analogy

A limited calibration analogy helps make the measurement issue concrete. A sensor has a response range, bandwidth, sampling process, noise floor, and placement relative to the phenomenon. The fact that a quantity exists does not guarantee that a particular sensor registers it, and one sensor's response does not determine every other readout. A human organism should not be reduced to a sensor, but the analogy captures why conscious report, behavior, physiology, memory, affect, and choice are not interchangeable measurements.

The same analogy applies to viable ranges, but not as a single threshold. A system calibrated by history may reduce the future influence of one cue while increasing the influence of another, or show learning only in the contexts where the relevant association is retrieved. This does not imply that adversity generally improves regulation or that one person has a globally stronger 'breakwater.' It motivates explicit tests of discrimination, generalization, context, history-relative familiarity, and transfer. The analogy is explanatory only; the empirical claims must come from the relevant learning, stress, attention, and trajectory measures.

13.5 Selection without an inner selector; different lived worlds

Conscious life necessarily reflects selection relative to the much larger set of conditions capable of affecting the organism, but selection need not imply an inner selector. Distributed mechanisms can alter which information dominates, persists, or is integrated. Because bodily signals, external inputs, and internally generated contents can contribute differently across states, the same external environment can accompany substantially different lived trajectories. This is a consequence of established attention, interoception, and self-generated-thought literatures rather than a novel three-world theory.

13.6 Cross-domain comparison as the potential payoff

Mechanistically distinct fields - including reward, memory, attention, interoception, source monitoring, action selection, and consciousness - can still inform the same inferential problem. Grey Matter's potential payoff is not to unify them under one cause, but to provide a common coordinate for asking whether a perturbation changed the lived trajectory, which relation was measured, and whether a mature local model already explains the effect.

13.7 Agency can include preserving a state

Agency is often easiest to see when an organism changes something, but change is not the only possible controlled outcome. A higher-order goal or valued context may favor maintaining an ongoing state despite an immediately aversive, distracting, or action-eliciting signal. Response inhibition provides mature precedent that internally guided control can suppress or pause prepotent action (Aron et al., 2014), while control-allocation theories provide domain-specific accounts of when effortful regulation is worth deploying (Shenhav et al., 2013). Grey Matter makes the narrower OPEN proposal that maintenance can sometimes be analyzed as one downstream relation of a lived trajectory. It does not infer agency from inactivity itself.

This matters for the breakwater intuition because a signal can remain phenomenally present without receiving automatic authority over the next transition. Fear can be experienced without immediate exit; bodily discomfort can be experienced while a chosen task continues; a valued quiet state can be preserved without adding stimulation. Those examples are not normative instructions. Harm, safety, consent, and context can make termination the appropriate controlled outcome. The theoretical point is only that phenomenality and termination are distinct relations.

13.8 Lived episodes can extend their causal history without defining phenomenality

A lived episode can matter later in at least two ways that should not be conflated. First, established encoding and consolidation processes may make some aspects of the episode more accessible in the future. Second, later retrieval or reflection can itself become a new episode that changes interpretation, association, expectation, or subsequent learning. Emotional-memory, reward-memory, reappraisal, and retrieval-learning literatures provide mechanism-specific examples (McGaugh, 2015; Miendlarzewska et al., 2016; Ochsner & Gross, 2005; Chan et al., 2018; Kensinger & Ford, 2020). None licenses reconstruction of the original phenomenology from later memory alone.

14. Collapse and falsification conditions

The framework should shrink whenever a simpler account succeeds. The following conditions are not peripheral caveats; they determine whether containment deserves to remain in the vocabulary.

Dimensional collapse: if boundedness, stability, and coherence cannot be dissociated under sensitive manipulations and converge on one latent factor, reduce them to the lower-dimensional structure supported by data.

Prior-art collapse: if dynamic-thought, active-inference, or neural state-space models already provide the same perturbation-to-lived-trajectory identification architecture, classify the residual contribution as SUBSUMED and narrow or abandon Grey Matter accordingly.

Lower-dimensional victory: if arousal, salience, surprise, precision or volatility, control demand, source monitoring, sensory gating, automaticity, or another mature construct fully explains the declared effect, use that construct and drop containment for that effect.

Relation-audit redundancy: if relation typing never changes an inference, prediction, or interpretation beyond ordinary construct-specific good practice, the non-entailment layer adds no residual contribution.

Perturbation redundancy: if rate, duration, familiarity, source, or other proposed variables add no reproducible information beyond endpoint magnitude, explicit predictability, or established alternatives, remove them from the scaffold.

Phenomenology failure: if the chosen measurements cannot support any defensible bridge to the lived trajectory, the study can report behavior, access, physiology, or neural dynamics but cannot claim containment of experience.

Post hoc rescue failure: if containment is invoked only after results appear and can explain any direction of change, the construct is unfalsifiable and should be abandoned.

Causal-role redundancy: if state maintenance or termination is fully explained by established response-inhibition, avoidance, reward, action-selection, or control-allocation models and relation typing adds no containment-relevant inference, use those models and drop the higher-level causal-role claim for that domain.

Calibration reduction: if repeated experience produces only global habituation, or if cue- and context-specific changes are fully captured by inhibitory learning, prediction error, safety learning, discrimination, reinforcement learning, or another mature account, those mechanisms win. Grey Matter cannot relabel their residual as selective containment.

Retention reduction: if later accessibility is fully explained by established encoding, consolidation, retrieval, reward-memory, or emotional-memory mechanisms, retention should be described in those terms. Its presence does not establish a Grey Matter-specific causal process, and its absence does not undermine prior phenomenality.

Reflection null: if later reflection changes only report or narrative and produces no reproducible change in future perturbation response, expectation, association, or retrieval structure, the proposed reflection-mediated reorganization should be narrowed to the effects that are actually measured.

Episode-separation failure: if the design cannot distinguish evidence about the original lived episode from evidence generated during later recollection or reflection, it cannot claim that the later measure reveals the original phenomenology or its original causal role.

15. Scope, branding, and metaphor discipline

Grey Matter is branding, not a scientific construct. The phrase "what's real / what's lived" frames a measurement problem and makes no metaphysical claim that unexperienced conditions are unreal. The scientific object is the organization of lived experience under perturbation and the inferential architecture required to identify it.

The breakwater remains a pedagogical analogy only. It is useful because a breakwater changes how variable input affects a protected region without eliminating the environment. It is incomplete because containment can involve incorporation, integration, contextualization, source attribution, recurrent stabilization, selective attention, or rapid updating rather than one-way attenuation. No formal definition or prediction depends on the metaphor.

The selective-calibration discussion likewise does not prescribe remaining in discomfort. Exposure and inhibitory-learning findings are used only as precedent that continued contact can support expectancy updating, discrimination, and context-dependent retrieval under controlled conditions. They do not show that all distress should be endured, that harmful environments should be maintained, or that reduced avoidance is universally adaptive. Termination can itself be the appropriate controlled response.

The framework does not claim that consciousness is necessary for all behavior, that report equals phenomenality, that memory proves prior experience, that a lived state necessarily causes later behavior, that pathology is a containment failure by definition, or that greater constraint is always better. It does not infer phenomenality from nonhuman behavior. Personal observations can motivate questions but are not evidence. It does not introduce generic weighting, control-leverage, consequence-profile, body/world/model, or memory-value scalars as new constructs. Established body/interoceptive, external-attention, internal-attention, self-generated-thought, reward, control, inhibition, learning, automaticity, and memory constructs retain their own names and explanatory priority. Reflection is treated as a later episode, not part of the original event, and later recollection is never assumed to reproduce original phenomenology perfectly.

Reference-architecture breadth does not make the framework indefinitely elastic. Progressive constraint is prospective: constraints, target relations, and comparison models should be declared before the critical inference rather than introduced post hoc to rescue a failed claim. If a candidate property fails to dissociate, a perturbation class adds no information, a downstream branch proves domain-specific, or a mature local theory fully explains an effect, the corresponding Grey Matter component must be merged, narrowed, or removed under the existing loss rules. The stable commitment is the identification discipline around perturbation, lived-trajectory property, relation-typed evidence, justified phenomenology bridges, and explicit loss to better explanations - not preservation of any particular lower-level mechanism or decomposition.

16. Conclusion

Conscious experience unfolds within an organism that changes continuously. Existing science already shows that thought can be constrained or variable, neural states associated with conscious experience occupy structured and dynamically accessible state spaces, biological regulation can maintain stability through change, attention, predictive precision, and active-inference models formalize selective influence and adaptive updating, action and motivation are not identical, organized behavior can become partly automatic, non-action can sometimes be actively controlled, learning can be cue- and context-specific, memory is not a transparent record of prior experience, and retrieval or reappraisal can alter later processing. Grey Matter claims none of those component findings as new.

Those precedents remain essential because they establish the measurement problem: outward behavior, report or access, affect, motivation, maintenance, and later recollection do not transparently identify the same relation to lived experience. That is why a containment hypothesis requires relation typing rather than why established supporting science should be removed.

The residual proposal is therefore narrow but concrete. Study the lived trajectory as an explicit perturbation-response target. Parameterize the perturbation. Distinguish boundedness, stability, and coherence only where data justify the distinction. Define lived by phenomenal presence alone, then measure any later role separately: maintenance or termination, learning and selective calibration, retention and future accessibility, or reflection-mediated reorganization. Use established interoception, attention, self-generated-thought, gating, precision, control, inhibition, learning, reward, memory, and source-monitoring constructs where they apply. Type every observation by the

relation it actually measures. Then require the higher-level containment construct to compete against mature lower-dimensional and state-space explanations.

On this account, consciousness is neither defined as an inert display nor granted automatic causal authority. A lived state is phenomenally present by definition. What that presence subsequently does is an empirical profile. It may help preserve the current state, contribute to a decision to terminate it, support new discrimination through continued sampling, be preferentially retained, or become the object of later reflection; it may also leave little detectable downstream consequence. The framework earns value only if keeping those relations separate improves inference or prediction.

As a reference architecture, this baseline is deliberately minimally precommitted. Later work can impose explicit constraints, test how the same typed perturbation-to-trajectory relations change, and retain only the components that survive mature competitors. This allows the architecture to become more specific without pretending that its final biological, environmental, developmental, or computational implementation was known in advance. It also leaves individual claims fully expendable: adaptation of the research program cannot be used to rescue a component after that component has failed its specified test.

Containment remains conditional and expendable. If established models fully explain the perturbation-to-trajectory effect and relation typing adds no inferential value, the term should disappear. If, however, preregistered comparisons reveal reproducible trajectory-level regularities across distinct mechanisms while preserving their differences, containment may earn a limited systems-level role: not as a hidden wall around consciousness, but as a name for an empirically identified stabilization problem whose evidence, mechanisms, and failure conditions are explicit.

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